

EXECUTIVE DATA REPORT

Title: Plug the Leak: Uncovering Superstore’s \$100,000 Discount Trap

Context: Sample Superstore Performance Analysis (n = 9,994 order lines)

Task: Day 4 Internship Data Storytelling Project

1. Executive Summary: High-Level Health Check

Superstore generates strong top-line revenue, but aggressive discounting erodes overall profitability. Nearly 1 in 5 orders sold results in an outright financial loss.

Key Performance Indicator	Metric Value	Benchmark / Insight
Total Sales	\$2.30M	High transaction volume across 9,994 order lines
Total Profit	\$286.40K	Net margin realized after accounting for all order lines
Overall Profit Margin	12.5%	Compressed margin due to deep promotional price cuts
Loss Rate %	18.7%	1 in 5 order lines sells below cost

Core Insight: The business model isn't failing on revenue; it is losing profit to unmonitored discount policies.

2. Leak Analysis: Profit Contribution by Sub-Category

Losses are heavily concentrated in a small selection of sub-categories rather than spread evenly across the product portfolio.

Primary Profit Drains (Negative Margin):

- Tables: -\$18,000 loss
- Bookcases: -\$3,000 loss
- Supplies: -\$1,000 loss

Top Profit Drivers (Positive Margin):

- Copiers: +\$56,000
- Phones: +\$45,000
- Accessories: +\$42,000
- Paper: +\$34,000

3. Evidence 1: Does Discounting Pay?

An evaluation of profit performance across four distinct discount tiers reveals that profitability collapses rapidly as discount depth increases.

Discount Band	Total Profit / Loss	Loss Rate %
0% Discount	+\$321K Profit	~0% Loss Rate
1% - 20%	+\$101K Profit	13.8% Loss Rate
21% - 40%	-\$36K Loss	90.2% Loss Rate
41%+	-\$100K Loss	100.0% Loss Rate

- **0%–20% Discount:** Highly profitable, capturing \$422K in cumulative profit.
- **21%–40% Discount:** Crosses into net negative territory (-\$36K loss), with 90.2% of lines turning unprofitable.
- **41%+ Discount:** Every single order line (100.0% loss rate) sells at a loss, accumulating \$100K in net losses.

4. Evidence 2: Category Vulnerability Matrix

Discount sensitivity varies across product categories, but deep discounts inflict severe damage on all lines.

Category	0% Discount	1%–20% Disc	21%–40% Disc	41%+ Disc	Overall Margin
Furniture	+22.7%	+5.1%	-18.3%	-70.9%	2.5%
Office Supplies	+29.5%	+16.5%	Break-even	-119.3%	17.0%
Technology	+34.0%	+14.7%	-8.8%	-50.8%	17.4%
Combined Avg	+29.5%	+11.9%	-15.3%	-77.4%	12.5%

- **Furniture** is the most sensitive category, turning unprofitable as early as the 21%–40% discount range.
- **Office Supplies** suffers catastrophic margin compression (-119.3%) under extreme discounting (41%+).

5. Strategic Recommendations & Financial Impact

By eliminating deep-discount policies and restricting promotional thresholds, Superstore can immediately recover \$100,000 in pure losses without selling a single additional unit.

Profit Bridge (Waterfall Projection):

Current Profit (\$286K) + Recovered Losses (\$100K) = Target Profit (\$386K)

3-Step Action Plan:

1. **Implement a Hard Cap on Discounts at 40%:** Discontinue all sales exceeding a 40% discount band where 100% of order lines operate at a loss.
2. **Set a Category Cap on Furniture at 30%:** Lower the discount threshold for Furniture specifically, as margins fall below zero starting in the 21%–40% window.
3. **Mandate Approval Workflows for Loss-Priced Lines:** Require regional manager sign-off for any commercial quote resulting in a negative profit margin.